

Weirs

The basic equations used to calculate weir flow are:

- Rectangular, Cipoletti, broad crested, and riser

$$Q = C_w L H^{1.5}$$

Where:

Q = Discharge over weir (cfs)

L = Length of the weir crest (ft)

H = Distance between water surface and the crest (ft)

C_w = Weir coefficient, typically 3.33

NOTE:

Hydraflow Hydrographs Extension uses the same weir equation for rectangular (sharp-crested weir with end contractions) and the Cipoletti weir (with no end contractions). Currently, there is not enough valid data available to support a unique equation for the weir with end contractions.

The following equation, supplied in HEC-22, attempts to adjust the weir length by subtracting 20% of H. However, by closer inspection, one can see that Q will eventually decrease to zero with increasing H.

$$Q = C_w (L - 0.2H) H^{1.5}$$

- V-notch

$$Q = 2.54 \tan\left(\frac{\theta}{2}\right) H^{2.5}$$

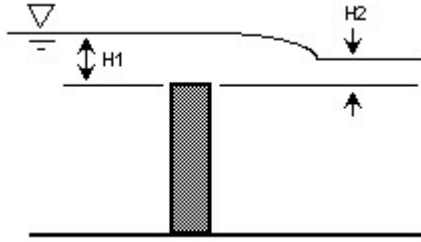
Where:

Q = discharge over weir (cfs)

θ = angle of v-notch (deg)

H = head on apex of v-notch (ft)

Rectangular, V-notch, and Cipoletti weirs are affected by submergence when the tailwater rises above the crest as follows.



This often occurs in multi-stage structures when the water surface in the riser (Riser HG) rises above the riser crest, due to the head produced by culvert A. As a result, the discharge over the weir is reduced. The equation for the reduction in flow is:

$$Q_s = Q_r \left(1 - \left(\frac{H_2}{H_1} \right)^{1.5} \right)^{0.385}$$

Where:

Q_s = submerged flow (cfs)

Q_r = unsubmerged flow from standard weir equations

H_1 = upstream head above crest (ft)

H_2 = downstream head above crest (ft)

NOTE:

Numbers that are adjusted for submergence have the suffix 's' in the stage-discharge table.

See Also

- [Weir Structures](#)